Structure–Prosody Integration in Predictive Processing: A human–multimodal LLM comparison in the visual world paradigm

Introduction. Predictive processing is widely regarded as a core mechanism for human language comprehension (Pickering & Garrod, 2013). Similarly, Large Language Models (LLMs) rely on a "next-token-prediction" objective that functionally parallels human anticipation (Brown et al., 2020). A further convergence between human and model systems is the use of "attention" to guide processing: eye movements in humans and attention weights in Multimodal LLMs (MLLMs, Vaswani et al., 2017). However, it remains unclear how humans and MLLMs *integrate structural and prosodic cues* to generate predictions in multimodal environments. The Visual World Paradigm (VWP) offers an ideal framework to examine these dynamics (Tanenhaus et al., 1995). This study places humans and MLLMs in an *identical VWP setting* to systematically compare their *offline behavioural outputs and online internal mechanisms*.

Method. We employed a *2 (Structure: DO vs. PO) × 3 (Stress: Neutral vs. Theme vs. Recipient)* design using six types of spoken Mandarin Chinese sentences (Fig.1, left). Each sentence was paired with a visual display containing four objects (Fig.1, right). Two experiments were conducted. **Experiment 1 (offline)** utilized a *cloze-in-VWP* task (Corps et al., 2022; Milburn et al., 2015), where auditory stimuli were truncated before the final target word. Both human participants (*N*=120) and the state-of-the-art MLLM, Qwen-2.5-omni (120 runs; Xu et al., 2025), selected the image that best completed the sentence and provided a continuation. **Experiment 2 (online)** employed a *standard VWP task*. Humans (*N*=59) passively listened to full sentences while eye movements were recorded. Simultaneously, Qwen was presented with the same audiovisual input and its attention weights to each picture were extracted.

Results. In the **offline task (Exp.1)**, DO sentences elicited more theme continuations than PO sentences ($p < .001$) in humans (Fig.2A), consistent with a Locality bias (Harris & Carlson, 2016). Stress modulated this bias—theme-stress increased theme continuations and recipient-stress reduced them (*ps* <.001). Critically, Stress had a larger impact when it mismatched the sentence-final position ($p < .01$), supporting competitive integration between structure and prosody. *Qwen closely matched the human pattern* in its continuation choices (Fig.2B). However, the **online task (Exp. 2)** revealed a dissociation. We utilized Linear Mixed-Effects (LME) for window-based analysis and Generalized Additive Mixed-Effects Models (GAMM) for time-course analysis. For humans (Fig.3A), *structural effects emerged early* (during the "Pause" window) and persisted (p =.005), while *stress effects emerged later* (during the "Connective" window, p =.001). Time-course analyses further suggested that prosody redirected looks primarily when it conflicted with the structure-based default (*ps* <.05). In contrast, Qwen's attention weights exhibited *no reliable anticipatory* differences by Structure or Prosody, and its time course *did not mirror human predictive looking* (Fig.3B).

Discussion. The results highlight a critical distinction between human minds and artificial machines in language processing. **Offline**, **both humans and MLLMs effectively integrate structural and prosodic cues,** demonstrating comparable multimodal capabilities in generated output. **However, online measures reveal fundamental mechanistic differences**. Human prediction is robust, hierarchical and dynamic (*structure earlier; prosody later*). In contrast, the *MLLM lacks the anticipatory "predictive looking" mechanism* observed in humans, suggesting that while the model's final output mimics human behaviour, its internal attention dynamics do not yet replicate the time-course of human predictive processing. These findings sharpen how "prediction" should be compared across minds and machines and motivate **caution in equating computational attention with human attention** (Jain & Wallace, 2019).

Design: 2 (Structure: DO vs. PO) x 3 (Stress: Neutral vs. Theme vs. Recipient)

- **DO structure:**

他	送	给	了	女孩	蛋糕,	而不是	男孩	/	鲜花。
3SG-he	V-give	DAT-to	PFV-le	Recipient-girl	Theme-cake	Conn-but not	Recipient-boy		Theme-flower
				mentioned recipient	mentioned theme		Alternative recipient		Alternative theme

("He gave the girl the cake, but not the boy/flower.")

- **PO structure:**

他	送	了	蛋糕	给	女孩,	而不是	男孩	/	鲜花。
3SG-he	V-give	PFV-le	Theme-cake	DAT-to	Recipient-girl	Conn-but not	Recipient-boy		Theme-flower
			mentioned theme		mentioned recipient		Alternative recipient		Alternative theme

("He gave the cake to the girl, but not the boy/flower.")

	DO structure	PO structure
Neutral	He gave the girl the cake, but not the boy/flower.	He gave the cake to the girl, but not the boy/flower.
Theme	He gave the girl the **cake**, but not the flower.	He gave the **cake** to the girl, but not the flower.
Recipient	He gave the **girl** the cake, but not the boy.	He gave the cake to the **girl**, but not the boy.

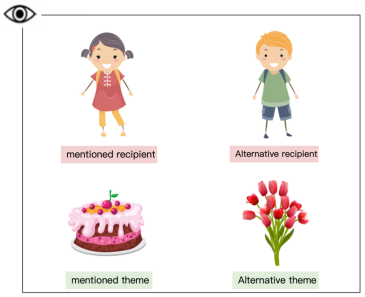

Figure 1. *Experimental Design and Visual Display*. Left: Spoken sentence design. Right: Example of visual display.

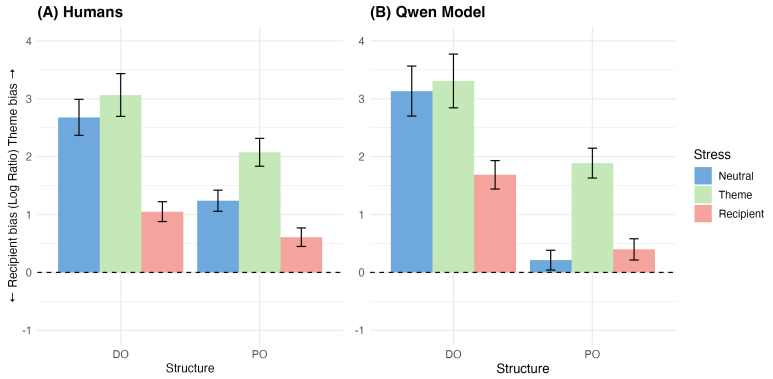

Figure 2. *Offline prediction bias in exp. 1(cloze-in-VWP)*. Bars show the smoothed log ratio: log((alternative Theme + 0.5)/(alternative Recipient + 0.5)): Values above 0 reflect a Theme bias, and values below 0 reflect a Recipient bias. Error bars indicate 95% CI.

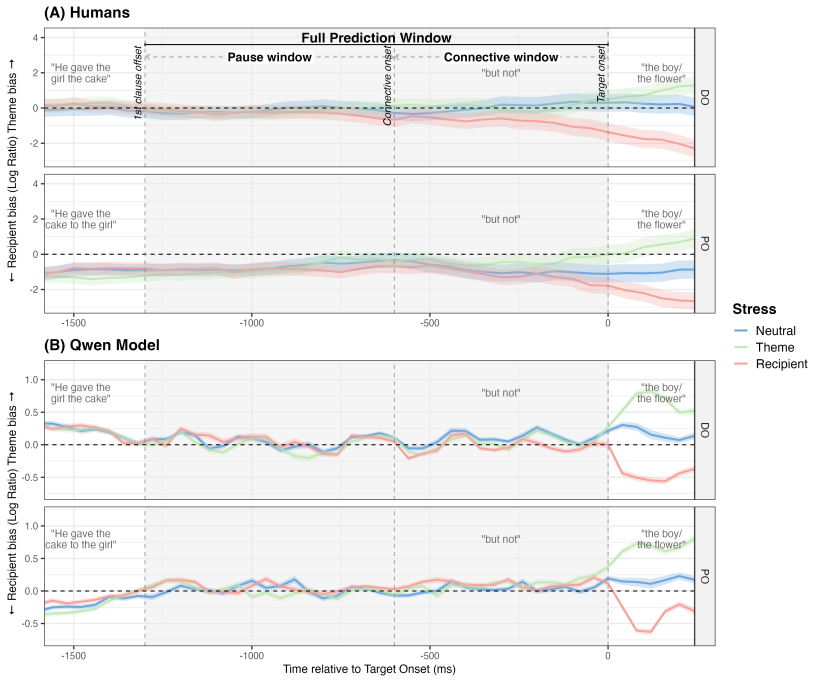

Figure 3. *Online predictive bias time-course in exp. 2 (standard VWP)*. Data are aligned to target onset (0 ms). The y-axis shows the log-ratio bias, derived from fixation proportions (Humans) and attention weights (Model). Values above 0 reflect a Theme bias, and values below 0 reflect a Recipient bias. The grey background region (-1300 to 0 ms) marks the prediction window (after the first clause and before the target word). Shaded areas represent 95% CIs.

Reference

- Brown, T. (2020). *NeurIPS*, 33, 1877–1901.
- Corps, R. E., et al. (2022). *JML*, 122, 104298.
- Harris, J. A., & Carlson, K. (2016). *QJEP*, 69, 1278–1301.
- Jain, S., & Wallace, B. C. (2019). Attention is not Explanation. *arXiv*:1902.10186.
- Milburn, E., et al. (2015). *Lang. Cogn. Neurosci.*, 31, 536.
- Pickering, M. J., & Garrod, S. (2013). *BBS*, 36, 329–347.
- Tanenhaus, M. K., et al. (1995). *Science*, 268, 1632–1634.
- Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS*, 30.
- Xu, J., et al. (2025). Qwen2.5-omni technical report. arXiv: 2503.20215v1.